9I Construct the Burrows-Wheeler Transform of a String

Burrows-Wheeler Transform Construction Problem
Construct the Burrows-Wheeler transform of a string.

Input: A string *Text*.
Output: BWT(*Text*).

```
$BANANAS
ANANAS$B
ANAS$BAN
AS$BANAN
BANANAS$
NANAS$BA
NAS$BANA
S$BANANA
```

Formatting

Input: A string *Text*.
Output: A string *Transform* representing BWT(*Text*).

Constraints

- The length of *Text* will be between 1 and 10^3.

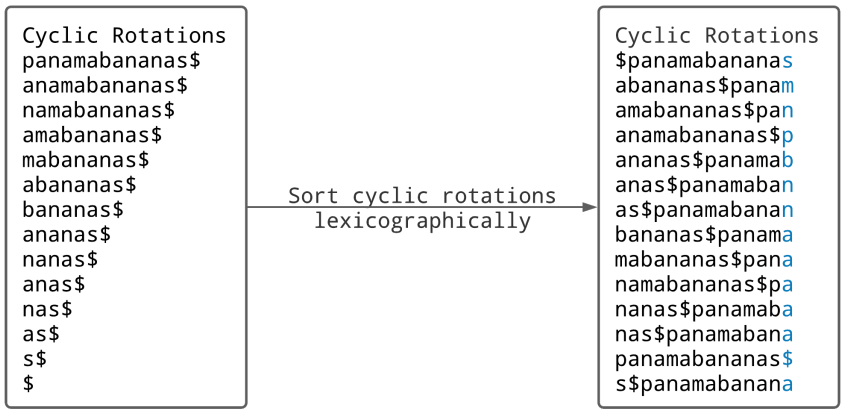
Test Cases

Case 1

Description: The sample dataset is not actually run on your code.
Input:
`panamabananas`

Output:
`smnpbnnaaaaa$a`

Figure:



Shown above is a general (and inefficient) construction process for the Burrows-Wheeler Transform for the string `panamabananas`. We first generate all cyclic rotations of *Text* before sorting them lexicographically to build a Burrows-Wheeler matrix. Lastly, we output the last column of the Burrows-Wheeler matrix as the Burrows-Wheeler Transform of *Text*. This process is very similar to the process of generating a suffix array.

Case 2

Description: There are repeats in *Text*.

Input:
```
AATCAATC
```

Output:
```
CC$AATTAA
```

Case 3

Description: *Text* is made up of only one character.

Input:
```
AAAAAAAAAA$
```

Output:
```
AAAAAAAAAA$
```

Case 4

Description: *Text* is palindromic or has substrings that are palindromic.

Input:
```
GAGCAT$
```

Output:
```
TGCG$AA
```

Case 5

Description: A larger dataset of the same size as that provided by the randomized autograder.